# Improper Integrals, Limits at Infinity & L'Hopital's Rule

MATH 146 – Calculus II

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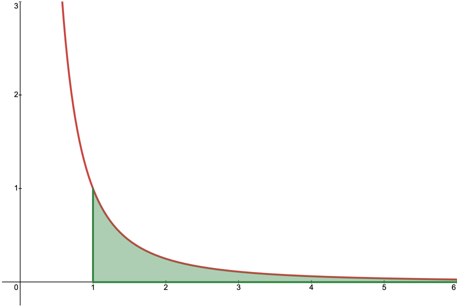
# Improper Integrals

**Example:** Find (if possible the area contained between the $x$-axis and the graph of $f(x) = \dfrac{1}{x^2}$ for $x \geq 1$.




Can we use an integral to find an area like this (if possible)? What would the bounds of integration be here?

# An Issue with the Definition of the Definite Integral

Recall:

## Definite Integral

Given a partition $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ of an interval $[a, b]$ and any choice of $x_i^*$ in $[x_{i-1}, x_i]$, if $f(x)$ is a function on $[a, b]$, then the **definite integral** on $[a, b]$ is:

$$\int_a^b f(x)\, dx = \lim_{\Delta x \to 0} f(x_i^*)\, \Delta x_i$$

How do we adapt this definition to deal with potentially infinite integrals? We'll have to consider *limits at infinity*.

# Important Limits at Infinity

- $\lim_{x\to\infty} \frac{1}{x^p} = 0$, if and only if $p > 0$.
- $\lim_{x\to\infty} \ln(x) = \infty$.
- $\lim_{x\to\infty} e^x = \infty$.
- $\lim_{x\to-\infty} e^x = 0$.
- $\lim_{x\to\infty} e^{-x} = 0$.
- $\lim_{x\to\infty} \tan^{-1}(x) = \frac{\pi}{2}$.
- $\lim_{x\to-\infty} \tan^{-1}(x) = -\frac{\pi}{2}$.

# Limits of Indeterminate Form

Suppose you have a limit such as the following, which come out to be in *indeterminate form*, i.e. the form $\dfrac{0}{0}$ or $\frac{\infty}{\infty}$.

$$\lim_{x\to 0}\frac{\sin(x)}{x}=\frac{0}{0} \qquad\qquad \lim_{x\to\infty}\frac{\ln(x)}{x^2}=\frac{\infty}{\infty}$$

To solve these kinds of integrals, we need to use **L'Hopital's Rule**.

# L'Hopital's Rule

## L'Hopital's Rule

Suppose $f$ and $g$ are differentiable functions over an open interval containing $a$, except possibly at $a$. If $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$ or $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \pm\infty$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

Assuming the limit on the right exists, or is $\infty$ or $-\infty$. This result also holds if we are considering one-sided limits, or if $a = \infty$ or $-\infty$.

# Example: L'Hopital's Rule

**Example:** Solve $\lim\limits_{x \to 0} \dfrac{\sin(x)}{x}$.

**Example:** Solve $\lim\limits_{x \to \infty} \dfrac{\ln(x)}{x^2}$.

# Definition of an Improper Integral

## Definite Integral

Suppose $f(x)$ is continuous. Then:

1. $\displaystyle\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx.$
2. $\displaystyle\int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx.$
3. $\displaystyle\int_{-\infty}^\infty f(x)\,dx = \lim_{a\to-\infty}\int_a^c f(x)\,dx + \lim_{b\to\infty}\int_c^b f(x)\,dx.$

This holds ***only if*** this limit exists. If it does, we say the improper integral **converges**. Otherwise, the improper integral **diverges**.

# Example: Improper Integrals

**Example:** If $\int_1^\infty \frac{1}{x^2}\, dx$ converges, find the value. Otherwise, say it diverges.

**Example:** If $\int_1^\infty \frac{1}{\sqrt{x}}\, dx$ converges, find the value. Otherwise, say it diverges.

**Question:** For what values of $p$ does $\int_1^\infty \frac{1}{x^p}\, dx$?

# A Comparison Theorem for Improper Integrals

## A Comparison Theorem

Let $f(x)$ and $g(x)$ be continuous over $[a, \infty)$. Assume that $0 \leq f(x) \leq g(x)$ for $x \geq a$.

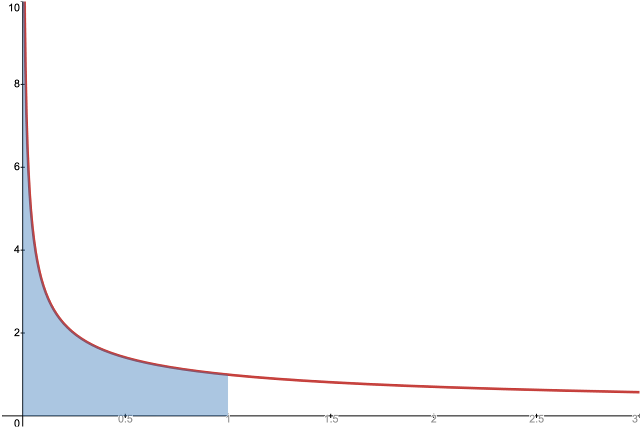
1. If $\displaystyle\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx = \infty$, then $\displaystyle\int_a^\infty g(x)\,dx = \lim_{t\to\infty} \int_a^t g(x)\,dx = \infty$.
2. If $\displaystyle\int_a^\infty g(x)\,dx = \lim_{t\to\infty} \int_a^t g(x)\,dx = L$, where $L$ is a real number, then $\displaystyle\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx = M$ for some real number $M \leq L$.

# Example: Comparison Theorem

**Example:** Determine whether or not the integral $\int_1^\infty \frac{1}{1+x^2}\,dx$ converges.

# Another Type of Improper Integral

**Example:** Evaluate $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,dx$. (Note that $f(x) = \dfrac{1}{x}$ has a vertical asymptote at $x = 0$.



# Improper Integrals with Asymptotes

1. Let $f(x)$ be continuous over $[a, b)$. Then:

$$\int_a^b f(x)\,dx = \lim_{t\to 0^-} \int_a^t f(x)\,dx.$$

2. Let $f(x)$ be continuous over $(a, b]$. Then:

$$\int_a^b f(x)\,dx = \lim_{t\to 0^+} \int_t^b f(x)\,dx.$$

In each case, if the limit exists, then the improper integral is said to converge. If the limit does not exist, then the improper integral is said to diverge.

# Improper Integrals with Asymptotes

3. If $f(x)$ is continuous over $[a, b]$ except at a point $c$ in $(a, b)$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

provided both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ converge. If *either* of these integrals diverges, then $\int_a^b f(x)\,dx$ diverges.

# Example: Improper Integral

**Example:** Evaluate $\displaystyle\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$, if possible.

**Example:** Find the values of $p$ such that $\displaystyle\int \frac{1}{x^p}$ converges.